

# Addition of Positive Integers Practice Worksheet

1.  $7 + 4$
2.  $23 + 15$
3.  $148 + 32$
4.  $467 + 285$
5.  $1,250 + 4,832$
6.  $56 + 189 + 43$
7.  $342 + 1,056 + 89$
8.  $2,415 + 3,890 + 764$
9. A library has 1,245 fiction books, 856 non-fiction books, and 340 reference books. What is the total number of books in the library?
10. A manufacturing plant produced 4,280 components in January, 5,125 components in February, and 3,950 components in March. How many total components were produced over the three-month period?

## Solutions Key: Addition of Positive Integers

1. **11**

*Explanation:* Basic single-digit addition:  $7 + 4 = 11$ .

2. **38**

*Explanation:* Align columns by place value:

$$\begin{array}{r} 23 \\ +15 \\ \hline 38 \end{array}$$

Add the ones digits ( $3 + 5 = 8$ ) and the tens digits ( $2 + 1 = 3$ ), resulting in 38.

3. **180**

*Explanation:* Align columns by place value:

$$\begin{array}{r} 1 \\ 148 \\ +32 \\ \hline 180 \end{array}$$

Ones:  $8 + 2 = 10$  (write 0, carry 1 to tens). Tens:  $4 + 3 + 1 = 8$ . Hundreds: 1. Total: 180.

4. **752**

*Explanation:* Column addition with regrouping:

$$\begin{array}{r} 11 \\ 467 \\ +285 \\ \hline 752 \end{array}$$

Ones:  $7 + 5 = 12$  (write 2, carry 1 to tens). Tens:  $6 + 8 + 1 = 15$  (write 5, carry 1 to hundreds).  
Hundreds:  $4 + 2 + 1 = 7$ . Total: 752.

5. **6,082**

*Explanation:* Multi-digit column layout:

$$\begin{array}{r} 1 \\ 1250 \\ +4832 \\ \hline 6082 \end{array}$$

Ones:  $0 + 2 = 2$ . Tens:  $5 + 3 = 8$ . Hundreds:  $2 + 8 = 10$  (write 0, carry 1 to thousands).  
Thousands:  $1 + 4 + 1 = 6$ . Total: 6,082.

6. **288**

*Explanation:* Aligning multiple terms vertically:

$$\begin{array}{r}
 1\ 1 \\
 5\ 6 \\
 1\ 8\ 9 \\
 +\ 4\ 3 \\
 \hline
 2\ 8\ 8
 \end{array}$$

Ones column:  $6 + 9 + 3 = 18$  (write 8, carry 1 to tens). Tens column:  $5 + 8 + 4 + 1 = 18$  (write 8, carry 1 to hundreds). Hundreds column:  $1 + 1 = 2$ . Total: 288.

7. **1,487**

*Explanation:* Aligning terms by place value:

$$\begin{array}{r}
 1\ 1 \\
 3\ 4\ 2 \\
 1\ 0\ 5\ 6 \\
 +\ 8\ 9 \\
 \hline
 1\ 4\ 8\ 7
 \end{array}$$

Ones:  $2 + 6 + 9 = 17$  (write 7, carry 1 to tens). Tens:  $4 + 5 + 8 + 1 = 18$  (write 8, carry 1 to hundreds). Hundreds:  $3 + 0 + 1 = 4$ . Thousands: 1. Total: 1,487.

8. **7,069**

*Explanation:* Standard column layout for three numbers:

$$\begin{array}{r}
 2\ 1 \\
 2\ 4\ 1\ 5 \\
 3\ 8\ 9\ 0 \\
 +\ 7\ 6\ 4 \\
 \hline
 7\ 0\ 6\ 9
 \end{array}$$

Ones:  $5 + 0 + 4 = 9$ . Tens:  $1 + 9 + 6 = 16$  (write 6, carry 1 to hundreds). Hundreds:  $4 + 8 + 7 + 1 = 20$  (write 0, carry 2 to thousands). Thousands:  $2 + 3 + 2 = 7$ . Total: 7,069.

9. **2,441 books**

*Explanation:* Sum all categories vertically:

$$\begin{array}{r}
 1\ 1\ 1 \\
 1\ 2\ 4\ 5 \\
 8\ 5\ 6 \\
 +\ 3\ 4\ 0 \\
 \hline
 2\ 4\ 4\ 1
 \end{array}$$

Ones:  $5 + 6 + 0 = 11$  (write 1, carry 1 to tens). Tens:  $4 + 5 + 4 + 1 = 14$  (write 4, carry 1 to hundreds). Hundreds:  $2 + 8 + 3 + 1 = 14$  (write 4, carry 1 to thousands). Thousands:  $1 + 1 = 2$ . Total: 2,441 books.

10. **13,355 components**

*Explanation:* Combine monthly production figures vertically:

$$\begin{array}{r}
 1\ 1\ 1 \\
 4\ 2\ 8\ 0 \\
 5\ 1\ 2\ 5 \\
 +\ 3\ 9\ 5\ 0 \\
 \hline
 \mathbf{13355}
 \end{array}$$

Ones:  $0 + 5 + 0 = 5$ . Tens:  $8 + 2 + 5 = 15$  (write 5, carry 1 to hundreds). Hundreds:  $2 + 1 + 9 + 1 = 13$  (write 3, carry 1 to thousands). Thousands:  $4 + 5 + 3 + 1 = 13$  (write 3, carry 1 to ten-thousands). Total: 13,355 components.